

Amir Mohammad Kharazi

✉ a.m.kharazi.ac@gmail.com ☎ 0991 177 21 10 🔗 <https://amkharazi.github.io/>

in amir-mohammad-kharazi 🌐 amkharazi

September 27, 2025

Research Interests

Deep Learning, Computer Vision, Transformers, Computational Neuroscience, Tensor Decomposition

Education

Tarbiat Modares University (TMU) 2022 – 2025

Master of Science (M.Sc.) in Computer Science

- GPA: 4.0 / 4.0 (Ranked 1st in the department)
- **Thesis:** Investigation of Tensor Methods in Vision Transformers

Amirkabir University of Technology (AUT) 2017 – 2021

Bachelor of Science (B.Sc.) in Computer Science

Research Experience

NMA - Impact Scholar Program 2024 – 2025

As part of the [Neuromatch Academy Impact Scholar Program](#), we conducted a comprehensive study comparing non-linear CEBRA and linear spatial filtering methods, evaluating their respective strengths and limitations when applied to ECoG data.

AI in Testing 2024 – Present

Led a pioneering team at [Digipay](#), developing test automation frameworks using Large Language Models (LLMs) and Computer Vision techniques for intelligent UI testing and interaction.

DeepTenLab 2023 – Present

Contributing member at [DeepTenLab](#), exploring tensor-based methods for computer vision and advanced neural architectures. Contributed to the development of modular PyTorch-based packages for layer-level substitution, enabling flexible experimentation beyond traditional architectures. Primary data modalities include images, but the work also extends to text, EEGs, and other types.

Teaching Experience

Computational Neuroscience Summer 2024, 2025
Neuromatch Academy (NMA)

Deep Learning Spring 2024
Tarbiat Modares University

Machine Learning Fall 2023
Tarbiat Modares University

Special Topics in Data Science Fall 2023
Tarbiat Modares University

Data Mining Spring 2021
Amirkabir University of Technology

Other Experience

Software Test Engineer

2024 – Present

Digipay

As a Software Test Engineer at [Digipay](#), one of the leading fintech companies in the region, I worked in an Agile development environment alongside cross-functional teams to integrate AI into software testing pipelines. My responsibilities included both automated and manual testing using state-of-the-art methodologies, as well as contributing to the design and maintenance of CI/CD pipelines aligned with modern industry standards.

Publications

*[Under Revision **Hierarchical Local-Global Tensor Layer(HLGTL) in Deep Convolutional Neural Networks***

2025

Expert Systems With Applications

Mansoor Rezghi, **Amir Mohammad Kharazi**, Mohammad Badzohreh

*[Submitted **ALGA: Input-Adaptive Local and Global Attention for Vision Transformers***

2025

IEEE Transactions on Pattern Analysis and Machine Intelligence

Mohammad Mahmoudabadi, **Amir Mohammad Kharazi**, Mansoor Rezghi

*[Micro Publication **A Comparative Analysis of Spatial Filtering and CE-BRA Methodologies for Enhancing BCI Adaptability on ECoG Data***

2025

Zenodo

Dandawate, A., **Amir Mohammad Kharazi**, McPherson, L., Molina-Padilla, X., Srinivasan, J., Teyavongsak, J., Fahimi Hnazaee, M.

doi.org/10.5281/zenodo.15319926

*[Under Preparation **VALTT: Vision All Layer Tensroized Transformer***

2025

Amir Mohammad Kharazi , Mansoor Rezghi,

Skills

Programming: C, R, Bash, Matlab, Java, Python,

Data Science: PyTorch, TensorFlow, Keras, torch_geometric, SciPy, MNE, Numpy, Pandas,

Technologies: Swagger, Kibana, MySQL, PostgreSQL, MongoDB, Redis, Robotframework, Selenium,

Web: HTML, CSS, JavaScript, Flask

Operating System: GNU, Linux,

Others: Git, CUDA, L^AT_EX

Languages

Persian: Mother Tongue

English: Proficient (**IELTS:** Overall 7.5 – L: 9.0, R: 7.0, W: 6.5, S: 7.0)

German: Basic

Hobbies

Computer

I enjoy playing video games and creating mods for them. My favorite games include Minecraft, Skyrim, and Dota2 .

Sports

I am an avid swimmer and also enjoy hiking and late-night running.

References

Dr. Mansoor Rezghi (rezghi@modares.ac.ir )

Associate professor at TMU Department of Computer Science.

Dr. Ali Rajaei (alirajaei@modares.ac.ir )

Assistant professor at TMU Department of Computer Science.

Neuromatch Academy

Recommendation letters available upon request.